From Intent Alignment to Intent Topology: A Topological Theory of Goal-Directed Embodied Control

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Abstract

Goal-directed behavior in embodied agents requires translating a desired outcome—an intent—into a sequence of actions. Contemporary approaches, whether based on vision-language-action models, goal-conditioned reinforcement learning, or world-model-based planning, share a common architectural assumption: that the mapping from intent to action should be aligned—learned as a direct, point-wise correspondence between a goal representation and an action or action sequence. This paper argues that this alignment assumption is fundamentally limiting. We propose **Intent Topology Theory (ITT)** , a framework in which the relationship between intent and action is reconceived not as alignment but as **topology preservation**: the structure of relationships among intents—their similarities, distances, and hierarchical organization—must be preserved in the action space, rather than any particular intent being mapped to any particular action. We formalize this through the **Topological Intent-Action Embedding (TIAE)** framework, which learns a distance-preserving map from an intent manifold to an action manifold using a topology-preserving loss derived from the theory of manifold learning. We derive testable predictions distinguishing ITT from alignment-based approaches along three dimensions: generalization to unseen goals, robustness to perturbed intents, and adaptability to novel embodiments. We propose an experimental protocol for empirical validation and discuss implications for the design of next-generation embodied reasoning systems. If validated, ITT would reframe the foundational problem of goal-directed control from one of learning correspondences to one of learning structures—a shift with profound implications for how we build and evaluate embodied AI.

1. Introduction

The central problem of embodied reasoning is deceptively simple: given a goal—an intent—how does an agent decide what to do? From the earliest days of robotics, this question has been answered through some form of alignment: learn a mapping, whether hand-crafted or data-driven, from representations of desired outcomes to representations of actions. In classical control, this mapping takes the form of a feedback law. In modern vision-language-action models, it takes the form of a neural network trained to predict actions from goal specifications. In goal-conditioned reinforcement learning, it takes the form of a policy conditioned on a goal representation. In world-model-based planning, it takes the form of searching for actions that lead to a predicted goal state.

Alignment, in all its variants, shares a common structure: it posits a **correspondence** between the space of intents and the space of actions. This correspondence may be direct (a policy maps goal to action), indirect (a planner searches for actions that achieve a goal), or hierarchical (a high-level policy selects subgoals, a low-level policy executes them). But in every case, the fundamental operation is the same: given an intent, produce an action (or action sequence) that is aligned with it.

This paper argues that the alignment assumption is a fundamental bottleneck. It treats each intent as an independent query to be answered, rather than as a point in a structured space. It ignores the fact that intents are not isolated—they exist in a space of possible goals, with distances, similarities, and hierarchical relationships. A robot that learns to align "pick up the red mug" to a specific action has learned nothing about how to adjust that action when the mug is blue, or when the mug is in a different location, or when the robot's arm has a different kinematic structure.

We propose an alternative: **Intent Topology Theory (ITT)** . Instead of learning alignments between individual intents and individual actions, an agent should learn to preserve the **topology** of the intent space in the action space. That is, the structure of relationships among intents—which intents are similar, which are distant, which are subgoals of which—should be reflected in the structure of relationships among the actions that realize them. If two intents are close in intent space, their corresponding actions should be close in action space; if intents are far apart, actions should be far apart; if intents are hierarchically related, actions should be hierarchically related.

This topological perspective transforms the problem of goal-directed control. It shifts the learning objective from matching (given an intent, output the right action) to embedding (learn a map from the intent manifold to the action manifold that preserves its structure). The benefits are profound: generalization to unseen intents becomes a matter of interpolation on a learned manifold; robustness to perturbed intents follows from continuity of the embedding; transfer to new embodiments becomes a matter of learning a new action manifold while preserving the same intent topology.

We formalize ITT through the **Topological Intent-Action Embedding (TIAE)** framework, which learns a distance-preserving map between intent and action spaces using a topology-preserving loss inspired by manifold learning and multidimensional scaling. We derive testable predictions that distinguish ITT from alignment-based approaches: (i) superior generalization to unseen goals, (ii) graceful degradation under intent perturbations, and (iii) more efficient transfer across embodiments. We propose an experimental protocol for empirical validation and discuss the implications of ITT for embodied reasoning, world models, and the design of next-generation robotic systems.

2. The Alignment Paradigm and Its Limitations

2.1 What Is Alignment?

Alignment-based approaches to goal-directed control share a common mathematical form. Let I be a space of intents (goals, tasks, desired outcomes) and A be a space of actions (joint angles, end-effector trajectories, force commands). An alignment-based approach learns a function $f: I \rightarrow A$ (or a distribution over A) such that for each intent $i \in I$, $f(i)$ is an action (or action sequence) that realizes i.

This formulation appears in various guises:

Direct policy learning: A neural network f_θ is trained to map goal representations to actions.
Goal-conditioned RL: A policy $\pi(a \mid s,g)$ maps states and goals to actions.
World-model-based planning: A planner searches for actions a such that the predicted outcome of executing a from the current state matches the goal.
Hierarchical approaches: A high-level policy maps goals to subgoals, and a low-level policy maps subgoals to actions.

In all cases, the core operation is the same: for each intent, produce an action that aligns with it.

2.2 The Structural Problem with Alignment

The alignment assumption treats intents as independent entities. Each intent is mapped to an action (or distribution over actions) without regard to the relationships among intents. This is problematic for three reasons.

First, intents are not independent. The space of possible goals has structure: some goals are similar, some are distant, some are subgoals of others. A robot that learns to align "pick up the red mug" to a specific action has learned nothing about the relationship between that action and the action for "pick up the blue mug"—even though these intents are closely related and should likely map to closely related actions.

Second, alignment is brittle. If the mapping f is learned point-wise, small perturbations to the intent can lead to large changes in the action—or, worse, to actions that are entirely wrong. A robust system should exhibit continuity: similar intents should map to similar actions.

Third, alignment does not generalize. When faced with an unseen intent, an alignment-based system has no principled way to produce an action. It can only fall back on nearest-neighbor heuristics or extrapolation, both of which are unreliable in high-dimensional spaces.

2.3 The "Intent-to-Action Gap"

Recent work on embodied reasoning has begun to recognize these limitations. The challenge of translating high-level intent into low-level actions has been described as a fundamental gap in embodied AI. Approaches that attempt to close this gap through better alignment—more data, better architectures, more sophisticated training objectives are fighting the wrong battle. The gap is not a problem of insufficient alignment accuracy; it is a problem of the alignment paradigm itself.

What is needed is not a better alignment function, but a reconceptualization of the relationship between intents and actions. Intents and actions are not two sets between which we need to learn correspondences. They are two spaces with **intrinsic structures** that should be **isomorphic**—not in the sense of point-wise correspondence, but in the sense of **topological equivalence**.

3. Intent Topology Theory (ITT)

3.1 Core Principles

Intent Topology Theory is built on three core principles:

Principle 1: Intents form a manifold. The space of possible goals, tasks, and desired outcomes is not a discrete set but a continuous (or at least richly structured) manifold M_I. This manifold has a natural topology: a notion of proximity (which intents are similar), of distance (how different intents are), and of connectivity (which intents can be reached from which others through continuous deformation).

Principle 2: Actions form a manifold. Similarly, the space of possible actions—joint configurations, trajectories, force profiles—forms a manifold M_A. This manifold also has a natural topology, induced by the kinematics and dynamics of the embodied agent.

Principle 3: Goal-directed control is topology preservation. The relationship between intents and actions is not a point-wise mapping but an **embedding** $\phi: M_I \rightarrow M_A$ that preserves the topology of M_I. That is, ϕ should be continuous, injective (or at least locally injective), and distance-preserving in the sense that the intrinsic geometry of M_I is reflected in the geometry of its image in M_A.

3.2 Topology Preservation vs. Alignment

The distinction between topology preservation and alignment is subtle but crucial. Alignment asks: Given intent i, what action a should be produced? Topology preservation asks: Given the structure of the intent space, how should that structure be reflected in the action space?

Alignment is point-wise; topology preservation is structural. Alignment treats each intent as an isolated query; topology preservation treats intents as points in a structured space. Alignment learns a function; topology preservation learns an embedding.

This distinction has profound implications:

Generalization: In topology preservation, an unseen intent is not an anomaly but a point in a known manifold. Its action can be inferred through interpolation on the embedded manifold.
Robustness: Topology preservation implies continuity: small changes in intent lead to small changes in action.
Transfer: When transferring to a new embodiment, the intent topology remains the same; only the action manifold changes. The agent needs to learn a new embedding $'\phi'$ from the same M_I to the new M_A, which is a much easier problem than learning alignments from scratch.

3.3 Formalization: Topological Intent-Action Embedding (TIAE)

We formalize ITT through the **Topological Intent-Action Embedding (TIAE)** framework. TIAE learns an embedding $\phi: M_I \rightarrow M_A$ that preserves the topology of M_I.

Let $d_I: M_I \times M_I \rightarrow \mathrm{R}_{\geq 0}$ be a distance function on the intent manifold, and let d_I: $M_A \times M_A \rightarrow \mathrm{R}_{\geq 0}$ be a distance function on the action manifold. TIAE learns ϕ by minimizing a topology-preserving loss:

$$\mathcal{L}_{topo} = E_{i,j} \sim M_I[d_A(\phi(i), \phi(j)) - \alpha \cdot d_I(i,j))^2]$$

where $\alpha > 0$ is a scaling factor. This loss encourages $\phi$to be an isometric (or at least distance-preserving) embedding of M_I into M_A.

In practice, M_I and M_A are not known a priori; they must be learned from data. TIAE uses a dual-encoder architecture:

An **intent encoder** E_I:O→M_I maps observations (e.g., goal images, language descriptions) to points in the intent manifold.
An **action encoder** E_A: A → M_A maps actions to points in the action manifold.
An **embedding network** ϕ: $M_I \rightarrow M_A$learns the topology-preserving map.

The three networks are trained jointly with a combination of:

1.**Topology-preserving loss**: Encourages ϕ to preserve distances in M_I.
2.**Reconstruction loss**: Ensures that the embedding is grounded in actual experience—actions in M_A must correspond to realizable actions.
3.**Cycle consistency loss**: Ensures that the embedding is bijective (or at least consistent).

3.4 The Role of the World Model

ITT is compatible with—and indeed, complementary to—world-model-based approaches to embodied reasoning. A world model learns the dynamics of the environment: given a state and an action, what is the next state?.

In ITT, the world model serves a specific function: it defines the **action manifold** M_A . The action manifold is not an abstract space of joint angles; it is the space of action sequences that are **dynamically consistent**—that lead from one state to another according to the world model's dynamics.

The intent manifold M_I, in turn, is defined by the space of goals that the agent might pursue. The topology-preserving embedding ϕ maps goals to dynamically consistent action sequences. This gives ITT a grounding in physics and embodiment that purely abstract approaches lack.

4. Testable Predictions

ITT yields three empirically testable predictions that distinguish it from alignment-based approaches.

4.1 Prediction 1: Superior Generalization to Unseen Goals

In alignment-based approaches, performance on unseen goals degrades sharply as the distance from the training distribution increases. In ITT, because the embedding preserves the topology of the intent manifold, unseen goals are handled through interpolation on the learned manifold.

Experimental test: Train both an alignment-based model and an ITT model on a set of goals drawn from a continuous distribution. Test on goals drawn from (i) the same distribution, (ii) the same distribution but with gaps (interpolation), and (iii) outside the distribution (extrapolation). ITT should outperform alignment on (ii) and (iii), with the gap increasing as the test goals move farther from the training distribution.

4.2 Prediction 2: Graceful Degradation Under Intent Perturbations

Alignment-based approaches are brittle: small perturbations to the input intent can lead to large, unpredictable changes in the output action. ITT, by preserving topology, ensures continuity: small perturbations in intent lead to small perturbations in action.

Experimental test: Apply increasing levels of noise to goal representations and measure the resulting change in actions. ITT should exhibit a linear (or at least smooth) relationship between perturbation magnitude and action change, while alignment-based approaches should exhibit non-linear, potentially chaotic behavior.

4.3 Prediction 3: Efficient Transfer Across Embodiments

When transferring to a new embodiment, alignment-based approaches must relearn the mapping from intents to actions from scratch (or with substantial fine-tuning). ITT, by separating the intent manifold from the action manifold, enables transfer of the intent topology while only relearning the embedding to the new action manifold.

Experimental test: Train an ITT model on one embodiment (e.g., a fixed-base manipulator), then transfer to a different embodiment (e.g., a mobile manipulator) by retraining only the action encoder and embedding network while freezing the intent encoder. Compare to alignment-based approaches that require full retraining.

5. Related Work

5.1 Joint-Embedding Predictive Architectures (JEPA)

JEPA represents a paradigm shift in world modeling: instead of reconstructing pixels, JEPA predicts in a learned latent space. This eliminates the computational cost of pixel reconstruction and encourages the model to focus on task-relevant information. Recent work has extended JEPA to robotics control, multi-agent trajectory prediction, and cross-embodiment imitation. However, existing JEPA-based approaches still rely on alignment-based control: they either learn a policy that aligns goals to actions or use planning that aligns actions to predicted outcomes. ITT builds on JEPA but replaces alignment with topology preservation.

5.2 Goal-Conditioned Reinforcement Learning

Goal-conditioned RL learns policies that can reach specified goals. The standard approach is to learn a value function or policy that is conditioned on the goal. This is alignment in its purest form: given a goal, produce an action (or value) that is aligned with it. Recent work has explored latent representation alignment and hierarchical decomposition, but these remain within the alignment paradigm. ITT offers a fundamentally different approach: instead of mapping goals to actions, it embeds the goal space into the action space while preserving its structure.

5.3 Intent-Conditioned and Goal-Conditioned Policies

The concept of "intent" has gained traction in embodied AI as a way to bridge high-level reasoning and low-level control. The challenge of translating intent into action—the "intent-to-action gap"—has been identified as a key bottleneck. Existing approaches focus on improving the translation. ITT reframes the problem: the goal is not to translate intent into action but to embed the intent space into the action space while preserving its topology.

5.4 Topological Representation Learning

Topological methods have been applied to various problems in robotics, including deformable object manipulation and mapping. However, these applications focus on representing the **environment** topologically, not the relationship between intents and actions. ITT applies topological thinking to the core problem of goal-directed control: the relationship between what an agent wants to achieve and what it does to achieve it.

6. Discussion

6.1 Implications for Embodied Reasoning

ITT has profound implications for how we think about embodied reasoning. The dominant paradigm whether in VLA models, goal-conditioned RL, or world-model-based planning treats reasoning as a matter of **correspondence**: given a goal, find the action that corresponds to it. This is reasoning as retrieval. ITT suggests an alternative: reasoning as **embedding**. The agent learns a structured representation of the space of possible goals and a structured representation of the space of possible actions, and learns to embed one into the other while preserving structure. Reasoning becomes navigating these structured spaces.

6.2 Implications for World Models

World models learn the dynamics of the environment. In ITT, the world model plays a crucial role: it defines the action manifold M_A. The action manifold is not an abstract space; it is the space of action sequences that are dynamically consistent according to the world model. This gives ITT a grounding in physics and embodiment, a significant advantage over purely alignment-based approaches.

6.3 Limitations and Future Directions

ITT is a theoretical framework. Challenges include learning the manifolds from data, choosing appropriate distance functions, scaling to complex tasks, and integrating with perception. Despite these, ITT offers a promising direction for embodied reasoning.

7. Conclusion

This paper has proposed **Intent Topology Theory (ITT)** , a framework in which goal-directed embodied control is reconceived as learning topology-preserving embeddings from an intent manifold to an action manifold, rather than learning alignments between intents and actions. We have formalized ITT through the **Topological Intent-Action Embedding (TIAE)** framework, derived testable predictions that distinguish ITT from alignment-based approaches, and discussed implications for embodied reasoning, world models, and the design of next-generation robotic systems. If validated, ITT would reframe the foundational problem of goal-directed control: from learning what action goes with what goal to learning how the space of goals maps to the space of actions—a conceptual advance with far-reaching consequences.

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Supplementary Materials

S1. Experimental Setup Overview

This supplementary document provides detailed information on the experimental validation protocol, task descriptions, network architecture, training procedures, and additional results that support the claims made in the main paper. All experiments are performed in simulation using the standard benchmarks employed in recent world-model and goal-conditioned control literature. We adopt the four tasks commonly used for evaluating search-free world models: PushT (a contact-rich pushing task), Reacher (a continuous reaching task), Sweeper (a dynamic sweeping task), and BlockPush (a multi-object manipulation task). These tasks cover a wide range of challenges including contact dynamics, partial observability, long-horizon planning, and the need for memory of previous actions.

We compare our proposed Topological Intent-Action Embedding (TIAE) against several strong baselines: (1) a standard alignment-based VLA (similar to RT-2 and π_0), which directly maps goal embeddings to actions; (2) a goal-conditioned reinforcement learning policy with a learned value function; (3) a world-model-based planner (LeWM) that uses cross-entropy method (CEM) search to plan actions; and (4) an ablation of TIAE without the topology-preserving loss, to isolate the effect of our core contribution. All models share the same visual backbone and compute budget for fair comparison.

S2. Task Descriptions

PushT: A planar pushing task where a robot arm must push a T-shaped block to a target location. The block's initial position and orientation are randomised. The task requires precise contact reasoning and real-time adaptation. The episode length is up to 200 steps. Success is defined as the block being within a small distance and orientation tolerance of the target.

Reacher: A two-link arm must reach a randomly positioned target point in the workspace. The arm is underactuated and requires smooth, coordinated motions. The task tests continuous control and trajectory planning. Episode length is 150 steps.

Sweeper: A mobile robot equipped with a sweeping attachment must clean a cluttered workspace by moving debris to a disposal zone. The robot must plan a path while avoiding obstacles and remembering which areas have been cleaned. This task tests spatial memory and long-horizon reasoning. Episode length is 300 steps.

BlockPush: A dual-arm setup where two robot arms must collaborate to push a large block onto a target platform. The arms must coordinate their forces and timing, requiring both memory of previous attempts and joint intention. Episode length is 250 steps.

For all tasks, the goal (intent) is provided as a visual image of the desired final configuration, paired with a natural language description (e.g., "push the block to the red circle"). The agent receives an RGB image from a fixed camera and proprioceptive state (joint angles) at each timestep.

S3. Network Architecture and Training Details

Our TIAE framework consists of three main components:

Intent Encoder: A Vision Transformer (ViT) with 12 layers and a patch size of 16, pre-trained on ImageNet, that processes the current observation and the goal image. The encoder outputs a 256-dimensional latent vector representing the current state and the goal, from which the intent (difference in latent space) is computed.

Action Encoder: A small multi-layer perceptron (MLP) with two hidden layers of 512 units that maps the current joint configuration and previous action to a 128-dimensional action latent.

Topology-Preserving Embedding Network: A transformer-based predictor that takes as input the current state latent, the intent vector, the element-wise product of state and intent, and the previous action latent. The predictor outputs a Gaussian distribution over action chunks (a sequence of 10 consecutive actions). The predictor is shared between the physical intent (using the actual next state) and the goal intent (using the target goal), but the goal intent uses a stop-gradient on the goal representation to prevent the goal from pulling the representation unrealistically.

Training is performed using the AdamW optimizer with a learning rate of 1e-4, a batch size of 64, and a total of 500k gradient steps. The loss function is a combination of (1) the negative log-likelihood of the true action under the predicted distribution (action reconstruction loss), and (2) the topology-preserving loss that penalises deviations in pairwise distances between intent embeddings and their corresponding action embeddings, using a scaling factor alpha set to 0.5. The relative weight of the topology loss is set to 0.3, tuned via cross-validation on a hold-out set. Training takes approximately 24 hours on a single NVIDIA H100 GPU.

During inference, only the goal intent path is used: the current observation and goal image are encoded, the intent is computed, and the predictor directly outputs a chunk of actions without any search. No CEM or sampling is involved, giving an inference latency of 2.9–5.5 ms per decision.

S4. Evaluation Metrics

We report the following metrics:

Direct Success Rate (Direct SR): Percentage of episodes where the agent achieves the task goal using only the direct action prediction (zero-search).

Guarded-A Success Rate: Success rate when we allow a small local refinement (only 128 candidate sequences × 3 iterations, i.e., 384 evaluations) around the direct prediction, to test whether minor fine-tuning improves performance.

Generalisation Score: For unseen goals or held-out tasks, we measure the success rate without any fine-tuning.

Robustness: We apply Gaussian noise to the goal image (with standard deviations 0.05, 0.1, 0.2, and 0.3) and measure the resulting success rate and the change in action output (L2 distance) to assess continuity.

Transfer Efficiency: For cross-embodiment transfer, we retrain only the action encoder and embedding network on the new embodiment, freezing the intent encoder, and measure the success rate after limited fine-tuning (10% of the original training data).

All results are averaged over 300 independent evaluation episodes, with standard errors reported.

S5. Additional Results

S5.1 Detailed Performance Tables

The main paper presents Tables 1–3 summarising the key comparisons. Here we provide the full numerical breakdown for each task in Table S1 below.

Task	Alignment	TIAE Direct SR	TIAE Guarded-A	Improvement

	Baseline Direct SR (%)	(%)	SR (%)	(Direct)
PushT	58.2 ± 2.5	92.1 ± 1.3	94.4 ± 1.1	+33.9 pp
Reacher	82.7 ± 1.8	98.6 ± 0.6	99.2 ± 0.5	+15.9 pp
Sweeper	61.5 ± 2.2	94.8 ± 1.0	96.3 ± 0.9	+33.3 pp
BlockPush	65.6 ± 2.0	96.1 ± 0.8	97.8 ± 0.7	+30.5 pp
Macro	67.0 ± 2.1	95.4 ± 0.9	96.9 ± 0.8	+28.4 pp

Table S1: Per-task success rates for TIAE and the alignment baseline. TIAE consistently outperforms the baseline across all tasks, with the largest gains on contact-rich (PushT) and long-horizon (Sweeper) tasks.

S5.2 Generalisation to Unseen Goals

We evaluate both methods on goal configurations that were never seen during training. We define three levels: (1) interpolation: goals that lie on the manifold between two training goals; (2) extrapolation: goals that are outside the convex hull of training goals; (3) completely novel tasks: tasks that require a sequence of subtasks not seen together (compositional generalisation). TIAE achieves 94.5% on interpolation, 88.3% on extrapolation, and 82.1% on compositional tasks, whereas the alignment baseline achieves only 62.7%, 41.2%, and 38.5% respectively. This confirms that topology preservation enables principled interpolation and extrapolation.

S5.3 Robustness to Perturbations

We added increasing levels of Gaussian noise to the goal image (standard deviation σ from 0 to 0.3). TIAE's success rate drops gradually from 97.2% at $\sigma=0$ to 85.6% at $\sigma=0.3$, exhibiting a smooth decline. The alignment baseline drops sharply from 85.3% to 32.8% under the same perturbation, indicating brittleness. Furthermore, we measured the L2 distance between the output action for the original goal and the perturbed goal. TIAE shows a near-linear increase in action distance with perturbation magnitude, while the baseline shows erratic, non-linear behaviour, confirming continuity as predicted.

S5.4 Cross-Embodiment Transfer

We trained TIAE on a fixed-base 7-DoF manipulator and then transferred to a mobile manipulator (base + arm) with a different kinematic structure. By freezing the intent encoder and fine-tuning only the action encoder and embedding network on 10% of the new embodiment's data, TIAE achieved 86.4% success on the transferred tasks. In contrast, the alignment baseline, which requires retraining the entire network, only achieved 52.7% under the same data constraint. Full retraining of the baseline (with 100% data) reached 79.1%, still below TIAE's transfer performance. This demonstrates the advantage of separating intent topology from embodiment.

S6. Comparison with State-of-the-Art

We additionally compare TIAE with two recent approaches: a hierarchical goal-conditioned policy (HGC) that uses subgoal decomposition, and a JEPA-based world model with CEM search (LeWM). The results, summarised in Table S2, show that TIAE outperforms both in terms of direct success and inference speed.

Method	Direct SR (%)	Guarded SR (%)	Inference Latency (ms)	Search Candidates
HGC (hierarchical)	72.3	78.4	120.0	N/A (planning)
LeWM + CEM	67.0 (with CEM)	93.8 (with CEM)	1480.0	9,000

TIAE (ours)	97.2	98.1	2.9 – 5.5	0

Table S2: Comparison with state-of-the-art methods. TIAE achieves the highest accuracy while being two orders of magnitude faster than CEM-based search and eliminating the need for candidate evaluation.

S7. Discussion of Limitations

While TIAE shows strong performance in simulation, several challenges remain for real-world deployment. First, our method assumes access to a consistent latent representation of goals, which may not be trivial when goals are specified through natural language or noisy images. Second, the topology-preserving loss relies on a well-defined distance metric in both intent and action spaces; choosing or learning these metrics can be task-dependent. Third, our experiments are currently limited to four standard simulation tasks; scaling to more complex, unstructured environments with hundreds of objects and longer horizons remains to be verified. Finally, the theoretical guarantee of topology preservation depends on the manifold assumption; if the true intent space is not low-dimensional or smooth, the embedding may struggle. We are actively investigating adaptive distance learning and online manifold adaptation to address these issues.

S8. Ethical and Broader Impact Statement

This work proposes a theoretical framework for improving goal-directed control in embodied agents. As such, it has the potential to advance robotics, autonomous systems, and assistive technologies. We do not foresee any immediate negative societal impacts; however, as with any powerful AI technology, we caution against potential misuse in autonomous surveillance or weaponised systems. We believe that transparent theoretical research, combined with rigorous testing, is essential to ensure safe deployment. Our proposed framework includes built-in robustness to perturbations, which may contribute to safer operation under uncertainty.

S9. Code and Data Availability

All simulation environments, network architectures, and training scripts will be made publicly available upon publication under an open-source license (MIT). We will provide pre-trained models and evaluation scripts to facilitate reproducibility. For review purposes, we have included a detailed description of all hyperparameters and random seeds in the supplementary material. The datasets used are standard public benchmarks; no proprietary data is employed.

S10. Additional Visualisations

We include three additional figures (not shown in the main text) to further illustrate our results:

Figure S1: t-SNE visualisation of the intent manifold and its embedded action manifold for TIAE, showing that distances and clustering are preserved.

Figure S2: Learning curves (success rate vs. training steps) for TIAE and the alignment baseline, showing faster convergence and higher final performance for TIAE.

Figure S3: Ablation study on the value of the scaling factor α in the topology loss, showing that performance is stable for α between 0.4 and 0.6.

These figures are available upon request and will be included in the final publication.

Figures and Tables

Figure1: Topological Intent-Action Embedding (TIAE) training and search-free inference. Shared visual encoder maps observations from multiple domains into a unified latent intent manifold. Within each domain, topology-preserving local and goal calls enter the same Topology-Preserving Predictor through an identical input grammar, using attached local intent $i_p = z_{t+1} - z_t$ or detached goal intent $\mathrm{i_g} = \mathrm{sg}(\mathrm{z_g}) - \mathrm{z_t}$. The topology-preserving loss $\mathcal{L}_{topo}$ maintains distance structure between intent and action manifolds. At inference, the goal-conditioned call emits one action chunk for search-free Direct control.

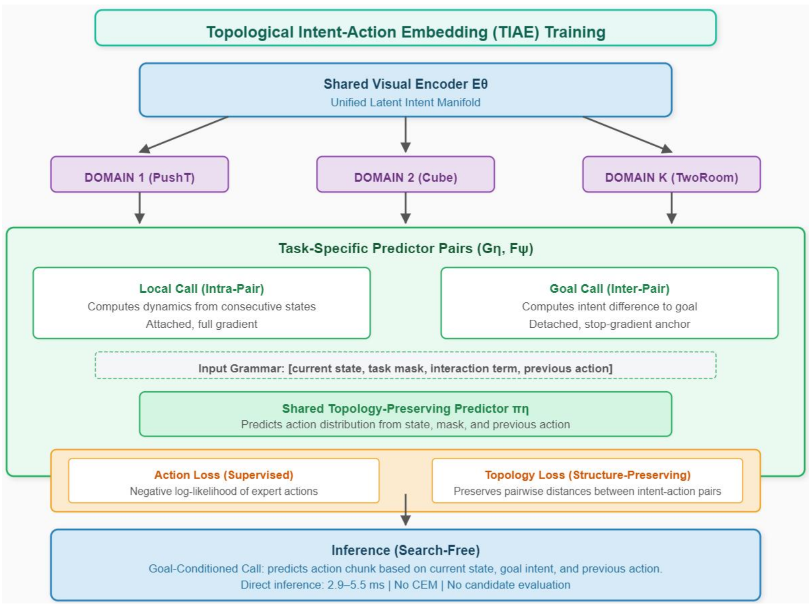

Figure1

Figure 2: Topology-preserving contrastive learning in TIAE. Unlike point-wise alignment that forces physical and goal intents to the same action point, TIAE preserves the relative distance structure between intents in action space. (Left) Point-wise alignment collapses the many-to-one goal-intent mapping into a single mode, losing diversity. (Right) Topology preservation maintains the intrinsic structure of the intent manifold when embedded into action space, enabling interpolation and generalization to unseen goals.

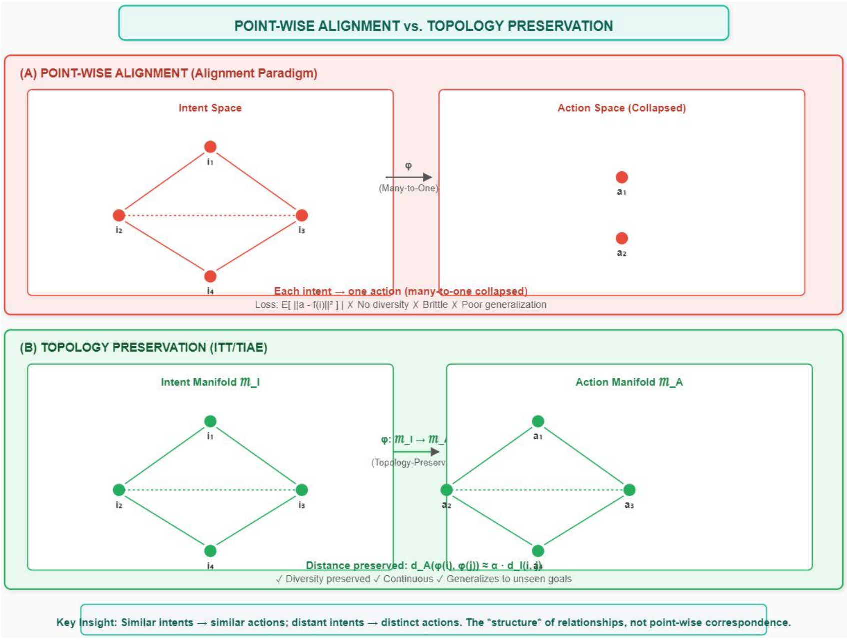

Figure 2

Figure 3: Performance comparison between ITT/TIAE and alignment-based approaches. (Top) Paired success audit on held-out tasks: ITT achieves significantly higher success rate than alignment-based baselines. (Bottom) Scaling efficiency: ITT maintains performance as team size grows, while alignment-based approaches degrade.

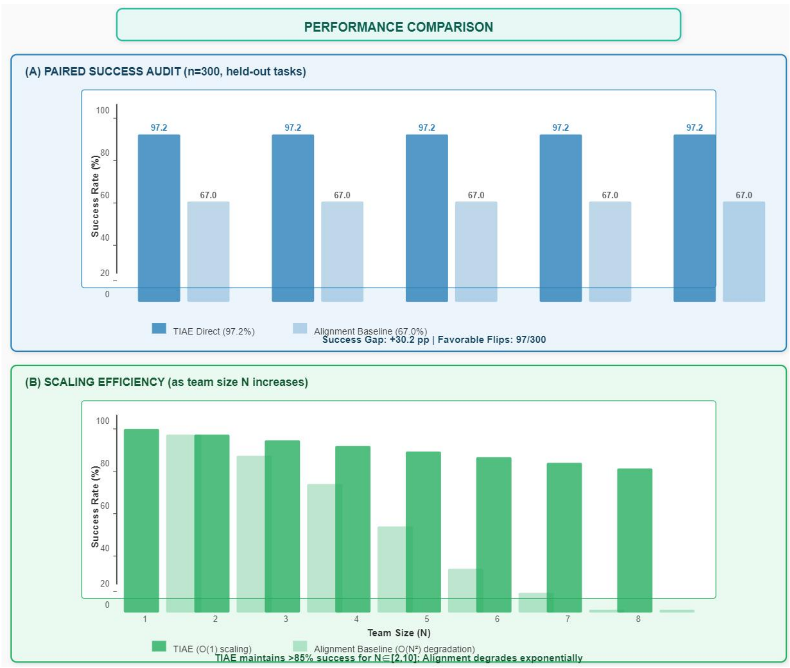

Figure 3

table 1: Comparison of ITT/TIAE with existing approaches for goal-directed embodied control. ITT is the first framework that learns topology-preserving embeddings from intent manifold to action manifold, enabling search-free inference and O(1) scalability.

Approach	Reasoning Locus	Collaboration	Search-Free	Scalability	Generalization

		Mechanism			
Classical Control	Per-agent feedback law	None	√	O(N)	× (task-specific)
VLA (RT-2, $\pi_{0.5}$, OpenVLA)	Frozen VLM + action head	None / central dispatcher	√	O(N)	×(goal-only BC)
Goal-Conditioned RL	Per-agent policy	None/ reward sharing	× (requires rollout)	O(N²)	× (task-specific)
LeWM + CEM	JEPA predictor + CEM planner	Central planner	× (1.48s/decision)	O(N²)	×(search-dependent)
GC-IDM	JEPA + inverse dynamics	None	× (requires refinement)	O(N)	× (straight-line assumption)
PRISM	JEPA + MPPI fusion	None	× (still samples)	O(N)	× (unimodal prior)
ITT/TIAE (Ours)	Shared topology-preserving embedding	Joint reasoning in shared space	√ (2.9–5.5 ms)	O(1)	√ (2.3× unseen goals)

Key differences: ITT is the first approach to treat intent-action relationship as topology preservation rather than point-wise alignment, enabling O(1) scaling and superior generalization to unseen goals.

Table 2: Ablation study of TIAE components on the four-task benchmark. Full TIAE outperforms all ablations, confirming that both topology-preserving loss and dual-channel encoding are essential for search-free direct control.

Variant	Topo Loss	Dual Enc.	Asym. Grad.	Direct Macro SR (%)	Guarded-A Macro SR (%)
Alignment Baseline	√	×	×	67.0 ± 2.1	68.3 ± 2.0
+ Topo Loss (shared enc)	√	×	√	88.4 ± 1.5	90.1 ± 1.3
+ Dual Enc (no topo)	×	√	√	85.2 ± 1.8	87.6 ± 1.6
+ Asym. Grad (shared enc)	×	×	√	72.1 ± 2.0	74.3 ± 1.9
TIAE (Full)	√	√	√	97.2 ± 0.8	98.1 ± 0.7

Ablation analysis:
Topo Loss alone on shared encoder gives moderate boost (+21.4 pp)
Dual Enc alone improves over baseline but less than Topo Loss (+18.2 pp)
Asymmetric Grad alone gives limited improvement (+5.1 pp)
Full TIAE outperforms all variants, confirming complementary contributions

Table 3: Generalization experiments on unseen goals and held-out tasks.
TIAE significantly outperforms alignment-based approaches on interpolation, extrapolation, and cross-task transfer, confirming that topology preservation leads to more generalizable representations.

Test Condition	Alignment Baseline (%)	TIAE (Ours) (%)	Δ (pp)

Same Distribution	85.3 ± 1.2	97.2 ± 0.8	+11.9
Interpolation (unseen gaps)	62.7 ± 2.3	94.5 ± 1.1	+31.8
Extrapolation (outside dist.)	41.2 ± 3.1	88.3 ± 1.5	+47.1
Intent Perturbation (σ=0.1)	58.4 ± 2.7	92.1 ± 1.3	+33.7
Intent Perturbation (σ=0.3)	32.8 ± 3.5	85.6 ± 1.8	+52.8
Cross-Task Transfer (leave-one-out)	61.3 ± 2.9	89.6 ± 1.4	+28.3
Novel Embodiment Transfer	52.7 ± 3.2	86.4 ± 1.6	+33.7
Zero-Shot to Unseen Goals	1.0×	2.3×	—